Task 2 – Exploratory Data Analysis & Business Intelligence

1. Introduction

Hello everyone.

In this Task 2, I worked on **Exploratory Data Analysis and Business Intelligence** using a sales dataset.

The main objective of this task was to understand the data, identify important patterns and trends, and answer business-related questions using **Python, SQL, and data visualizations**.

For this analysis, I used **Python with Pandas, NumPy, Matplotlib and Seaborn** for EDA, and **PostgreSQL** for answering business questions using SQL.

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2. Dataset Overview

First, I loaded the cleaned sales dataset into Jupyter Notebook.

The dataset contains **1,000 rows and 16 columns**.

Some important columns are:

- Order ID
- Order Date
- Customer ID
- Customer Name
- Age
- Gender
- City
- Product
- Category
- Quantity
- Unit Price
- Total Sales
- Year
- Month

I first checked the first few records using `df.head()`.

Then I checked the shape and data types of the dataset to understand its structure.

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3. Descriptive Statistics

Next, I performed descriptive statistical analysis using `describe()`.

From the analysis:

- The average customer age is approximately **41 years**.
- The average quantity per order is approximately **5.4 units**.
- The average unit price is around **25,487**.
- The average total sales value is approximately **139,399**.

The median age is around **41 years**, while the minimum age is **18** and the maximum age is **65**.

This gives us a basic understanding of the distribution of the numerical variables.

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4. Categorical Analysis

Next, I analyzed categorical variables.

For gender, the dataset contains:

- **511 Male customers**
- **489 Female customers**

So, the gender distribution is quite balanced.

I also analyzed customers by city.

The major cities in the dataset include **Patna, Kolkata, Mumbai, Hyderabad, Delhi, Bengaluru, Gaya and Pune**.

Patna has the highest number of records among the cities shown in the analysis.

I also checked the product distribution to understand which products occur most frequently in the dataset.

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5. Univariate Visualization

After descriptive statistics, I created visualizations.

First, I created an **Age Distribution histogram**.

This visualization helps us understand how customers are distributed across different age groups.

The ages are spread from approximately **18 to 65 years**, with a reasonable representation across the range.

Next, I created a **Quantity Distribution histogram**.

This shows how many units customers generally purchase in an order.

Finally, I created a **Total Sales Distribution histogram**.

From this visualization, we can observe that most sales transactions are concentrated in the lower-to-middle sales range, while very high-value transactions occur less frequently.

These visualizations make it easier to identify patterns that may not be obvious from raw data.

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6. SQL Business Questions

After completing EDA, I moved to PostgreSQL to answer business-related questions.

Before running the business queries, I first checked the sales table using:

`SELECT * FROM sales LIMIT 10;`

I also checked the column names and data types.

Since `Order_Date` was stored as text, I converted it into the proper **DATE** data type.

This is important because date-based filtering and analysis require a proper date format.

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7. Business Question 1 – Top 5 Products by Revenue

The first business question was:

Which are the top 5 products by revenue in the last 6 months?

I used `SUM(Total_Sales)` to calculate revenue and grouped the data by product.

I then sorted the result in descending order of revenue.

From the result, **Laptop and Book** are among the highest-revenue products.

This analysis can help the business identify its most valuable products and focus on inventory and marketing for those products.

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8. Business Question 2 – Category-wise Revenue

The second question was:

Which product categories generate the highest revenue?

For this, I grouped the data by `Category` and calculated the total sales using the `SUM()` function.

The result allows us to compare the revenue contribution of different categories such as Electronics, Education, Grocery and other categories.

This can help management understand which categories are contributing more to overall revenue.

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9. Business Question 3 – Top 10 Cities by Revenue

The third question was:

Which cities generate the highest revenue?

I grouped the sales data by city and calculated total revenue for each city.

Then I sorted the results in descending order and used `LIMIT 10` to display the top 10 cities.

This analysis helps the business identify its strongest geographical markets.

It can also support decisions related to regional marketing, inventory and expansion.

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10. Business Question 4 – Monthly Revenue Trend

The fourth question was:

How does revenue change month by month?

For this analysis, I grouped the data by **Year and Month** and calculated monthly revenue.

I also used a CASE statement to arrange the months in the correct January-to-December order.

This is important because alphabetical sorting would otherwise place months in the wrong order.

The monthly revenue analysis helps identify high-performing and low-performing periods and can be useful for understanding seasonal trends.

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11. Business Question 5 – Gender-wise Revenue

The fifth question was:

How much revenue is generated by each gender?

For this query, I calculated both:

- Total number of orders
- Total revenue

and grouped the results by Gender.

This helps us compare customer purchasing behavior across different gender groups.

The dataset has a nearly balanced distribution between male and female customers, so this comparison provides a useful overview of revenue contribution.

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12. Business Question 6 – Average Order Value

The sixth question was:

What is the average order value?

I calculated the average order value by dividing total sales by the number of distinct orders.

The SQL result gives an average order value of approximately **1,405,213.63** based on the current dataset and calculation.

This KPI is useful because it tells the business how much revenue is generated on average per order.

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13. Business Question 7 – Product-wise Quantity and Revenue

The final business question was:

Which products sell the most quantity and also generate high revenue?

For this analysis, I grouped the data by product and calculated:

- Total quantity sold
- Total revenue

This helps identify products that have both strong sales volume and strong revenue contribution.

A product with high quantity but low revenue may need a different strategy compared with a product having lower quantity but very high revenue.

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14. Business Insights

Based on my EDA and SQL analysis, I identified several useful insights.

First, the customer age distribution covers a broad range from 18 to 65 years.

Second, the gender distribution is almost balanced, with 511 male and 489 female records.

Third, different cities contribute differently to the overall sales, which means geographical performance can be an important business factor.

Fourth, some products contribute significantly more revenue than others, so product-level analysis is important for business decision-making.

Finally, monthly revenue analysis can help identify trends and seasonal changes in sales.

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15. Dashboard / KPI Mock-up

Based on these insights, I would create a business dashboard containing the following important KPIs:

- **Total Revenue**
- **Total Orders**
- **Average Order Value**

- **Total Quantity Sold**
- **Top 5 Products**
- **Revenue by Category**
- **Revenue by City**
- **Monthly Revenue Trend**
- **Gender-wise Revenue**

For visualization, I would use:

- KPI cards for important metrics
- Bar charts for product and city comparison
- A line chart for monthly revenue
- A pie or donut chart for category or gender distribution
- A histogram for numerical distributions

This dashboard would help management quickly understand business performance and make data-driven decisions.

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16. Conclusion

To conclude, this task helped me understand the complete flow of a basic data analytics project.

I started with **data inspection and descriptive statistics**, then performed **univariate analysis and visualizations**.

After that, I used **SQL to answer seven business questions** involving filtering, aggregation, grouping, sorting and date-based analysis.

Finally, I converted the findings into useful business insights and identified the key KPIs that can be included in a dashboard.

Overall, this task improved my practical understanding of **EDA, SQL, data visualization and Business Intelligence**.

Thank you.